

4. Experiment Design and Implementation

To examine the hypotheses outlined above—namely, the effects of different intensities of smart working (one day per week versus five days per week) on employee productivity, as well as the productivity patterns of women with varying family responsibilities—the following experimental design is proposed. The experiment was implemented within a single company for 6 months, where participants are randomly separated into 3 different groups. The control group C0 work still as usual, namely work five days per week at the office. The member of group T1 do smart working only one day per week and they can choose any day within the workweek as their smart working day. Employees in group T5 engage in smart working five days per week, amounting to approximately 20 days per month. First, the experiment uses survey questionnaires both before and after the experiment. The questions include the measures of self-reported productivity and supervisor-reported productivity. The final productivity measure was constructed as the average of these two assessments. Both measurements of the productivity include four same dimensions: output of production, efficiency at work, proactivity, and the capacity to comply with predetermined deadlines. In addition, participants in group T1 and T5 are asked to record the times of work interruptions they experienced each day. The average number of interruptions per day over the experimental period was then calculated to capture the frequency of work interruptions during smart working.

To ensure that the estimated effects of smart working reliably reflect causal relationships, the empirical models include several control variables that account for differences in employees' personal backgrounds, job characteristics, and home working conditions. Specifically, the control variables consist of age (capturing potential life-cycle effects), tenure (reflecting organizational experience), managerial position (as managers may follow distinct working patterns and productivity benchmarks), housing size (indicating the adequacy of home space), availability of an independent workspace at home (which affects concentration and interruptions), and commuting time (which may influence preferences for and efficiency under smart working). Including these controls helps mitigate omitted variable bias and improves the precision of the regression estimates.

To examine H1, following regression model is proposed:

$$Productivity_{it} = \alpha + \beta_1 D5_i + \beta_2 D1_i + \beta_3 Post_t + \beta_4 (D5_i \times Post_t) + \beta_5 (D1_i \times Post_t) + \gamma X_i + \mu_i + \varepsilon_i$$

Here, $D5_i$ is 1 when participant i belongs to T5, $D1_i$ is 1 when participant i belongs to T1. In control group C0, both $D5_i$ and $D1_i$ are 0. $Post_t$ is 1 if the outcome comes from the data after the experiment. X_i refers to control variables, while μ_i refers to individual fixed effects.

$D5_i \times Post_t$ is 1 if the participant i belongs to the group T5 and after the treatment. $D1_i \times Post_t$ is similar. β_4 and β_5 represents the difference-in-differences estimates of the causal effects of the five-days and one-day smart-working groups relative to the control condition. If $\beta_4 - \beta_5 > 0$ and it's significant, then H1 is true, which means it will achieve a higher productivity when employees do smart working every day rather than only one day every week. If women with children experience more work interruptions than women without children will be examined by the next regression model:

$$Interruption_{it} = \alpha + \beta Children_i + \delta SWIntensity_{it} + \gamma' X_i + \mu_i + \varepsilon_i$$

Here, $Interruption_{it}$ is defined as the average of the amounts of the interruptions per day recorded during smart working experiment. Since only women in group T1 and T5 are observed in this hypothesis, $Children_i = 1$ when individuals are mother, otherwise it's 0 when individuals are women without children. $SWIntensity_{it}$ represents the intensity of smart working, namely the number of days of smart working every week. This variable is added in this regression model because we have to ensure that the estimated coefficient on $Children_i$ is not confounded by differences in the number of smart-working days across workers. If $\beta > 0$ and $Children_i$ has a statistically significant effect on interruptions, then H2 is proved, namely mothers experience more interruptions than women without children.

Building on the first two hypotheses, the experiment enables a deeper examination of female employees with children. Since it is assumed that mothers will experience more work interruptions than women without children because of greater family responsibilities—for instance, the need to address urgent child-related matters—the analysis can further examine whether doing smart working five days per week results in lower productivity compared with those who conduct only one smart working day in a week among females with children. According to this, the third regression model is proposed:

$$Productivity_{it} = \alpha + \beta_1 D5_i + \beta_2 Post_t + \beta_3 (D5_i \times Post_t) + \gamma' X_i + \mu_i + \varepsilon_i$$

where individual i is mothers in group T1 and T5. $D5_i$ is 1 when individual i belongs to group T5, and $D5_i \times Post_t = 1$ when individual i belongs to T5 after the treatment. If $\beta_3 < 0$ and it's significant, then H3 is proved.